PETROBRAS

DR-ENGP-I-1.15

Revision 4

ENGINEERING GUIDELINES

Production Development and Technology

Offshore Production Facilities

Color coding

COLOR CODING

APPROVAL	SRGE/ESUP
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EXECUTION	SRGE/ESUP/EEA/EBET

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COLOR CODING

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1. SCOPE

This guideline is applicable to PETROBRAS offshore production installations.

2. OBJECTIVE

This technical specification establishes the minimum requirements conditions in the application of colors for identification of structures, equipment, piping and miscellaneous at offshore production units.

3. DEFINITION

TOP COLOR – final color, provided by the top coat or through natural color of substrate that cover the whole equipment, structure or piping.

4. NORMATIVE REFERENCE

- NR-26 - Sinalização de Segurança
- NR-37 - Segurança e Saúde em Plataformas de Petróleo
- NBR 6493 Emprego de cores para identificação de tubulações
- NBR 7195 Cores para Segurança
- Portaria nº 0256/GC5 de 13/05/2011– “Dispõe sobre as restrições relativas às implantações que possam afetar adversamente a segurança e a regularidade das operações aéreas, e dá outras providências”. Alterada pela Portaria COMAER 98/GC5, de 6 de março de 2012. Alterada pela Portaria COMAER 271/GC5, de 06 de junho de 2012.
- NORMAM 27 – Normas da Autoridade Marítima Para Homologação de Helipontos Instalados em Embarcações e em Plataformas Marítimas
- DR-ENGP-1.1 (Materiais de Tubulação para Instalações de Produção e Facilidades de Processo)
- Manual de sinalização PETROBRAS
- DR-ENGP-I-M-1.21 – Requisitos Gerais de Revestimentos

5. ABREVIATIONS

RC - Recommended Color

FRP – Fiber Reinforced Plastic

6. GENERAL REQUIREMENTS

6.1. All equipment, vessels and non-structural tanks shall be identified by corresponding Name, Function and TAG, painted visibly in Black (Munsell N 1) letters besides its nameplate.

6.2. The workplaces, machinery and equipment in general shall be painted with light colors, in order to enhance safety and ergonomics to facilitate operation, inspection and maintenance works, and to obtain higher energy and lighting efficiency.

6.3. Carbon Steel elements shall be coated according to the specific Project approved coating system and shall receive a top coat of painting according to the MUNSELL color system. Metalized carbon steel, aluminum and stainless steel shall be painted when specified in the Project documents.

6.4. The basic color is the top coat that covers the full length of piping and/or equipment.

6.5. Unless otherwise mentioned, the recommended top color for equipment shall be Light Green (MUNSELL notation 5 G 8/4). The colors green and white may be accepted.

7. DEVIATIONS AND EXCEPTIONS

7.1. For equipment marked as [RC] the corresponding color is the Recommended Color. In this case, any equipment manufactured with vendor standard color which deviates from this rule may be acceptable since approved by Petrobras.

7.2. Omitted cases or of conflict between referred documents shall be directed to PETROBRAS for approval.

8. COLOR CODING INFORMATION

8.1. The platform shall have charts containing the Color Coding of piping, fluids and corresponding line TAGs used, installed in strategic positions.

8.2. The standard of colors used in this technical specification and the correspondence with the MUNSELL notation are described in the following table.

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TABLE 1 –MUNSELL NOTATION

DENOMINATION		MUNSELL NOTATION
Black		N1
White		N 9.5
Aluminum		---------
Safety Red		5 R 4/14
Brown		2.5 YR 2/4
Safety Orange		2.5 YR 6/14
Safety Yellow		5 Y 8/12
Light Grey		N 6.5
Ice Grey		N 8
Petrobras Green		2.5 G 5/10
GREEN EMBLEM		2.5 G 3/4
Safety Blue		2.5 PB 4/10
Light Blue		2.5 PB 8/4
Light Cream		2.5 Y 9/4
Light Green		5 G 8/4

9. COLOR DEFINITION FOR STRUCTURES

9.1. External structures

9.1.1. All Steel Structures, Bulkheads, including:

a) Primary structure, including vertical columns and diagonal bracings: Light Grey (MUNSELL N 6.5)

b) Secondary structure – including penetration sleeves, and top of webs and flanges: Light Grey (MUNSELL N 6.5)

c) Bulkheads – including all stiffeners: Light Grey (MUNSELL N 6.5)

d) Deck floors in outdoors (decks, process areas): Light Grey (MUNSELL N 6.5)

9.1.2. Outfittings and specific structural elements:

a) Carbon Steel Deck and stairways grating: Light Grey (MUNSELL N 6.5)

b) Main Pipe rack: Light Grey (MUNSELL N 6.5)

c) Pipe, cable trays support: follows the adjacent deck, deck underside and/or bulkhead color.

d) Lighting fixtures pole: follows the supporting structure.

e) Equipment, vessels and tanks support foundation: follows the adjacent deck and/or bulkhead color.

f) Crane Pedestal: Light Grey (MUNSELL notation N 6.5).

g) Curb or salient structure where it is needed to call attention (risk to hit or flooring someone): Safety Yellow (MUNSELL notation 5 Y 8/12).

h) Front-ending of ladder steps: Safety Yellow (MUNSELL notation 5 Y 8/12).

i) Vertical or inclined Ladders, Stairways, Walkways and Platforms: Safety Yellow (MUNSELL notation 5 Y 8/12).

j) Safety cages, handrails, railing, guardrail, etc.: Safety Yellow (MUNSELL notation 5 Y 8/12).

NOTE: The steps of stairway that is part of the escape route, follows the escape route color.

k) Caisson: Black (MUNSELL notation N1)

l) Davit for Lifeboat and Rescue Boat: SafetyYellow (MUNSELL notation 5 Y 8/12)

m) Fenders (supply boat area): Black (MUNSELL notation N 1)

n) Flare structure: Aluminum

9.2. Semi-Submersible Hull (external):

9.2.1. Pontoons, columns, blister and bracings (immersed): Black (MUNSELL notation N 1).

9.2.2. Columns (emerged), Deck-box side and bottom (underside): Black (MUNSELL notation N 1).

9.3. FPSO hull:

9.3.1. External Hull: Black (MUNSELL notation N 1).

9.3.2. Bulwark External Surfaces: Black (MUNSELL notation N 1).

9.3.3. Bulwark Internal Surfaces: Ice Grey (MUNSELL notation N 8).

9.3.4. Turret, if is the case, (internal and external): Ice Grey (MUNSELL notation N 8).

9.4. Internal Structures Machinery, Utility and similar spaces, Pump rooms, Void spaces)

9.4.1. Bulkheads including all faces of secondary steel, stiffeners, penetrations, etc.: White (MUNSELL notation N 9.5).

9.4.2. Doors: White (MUNSELL notation N 9.5), see item Safety.

9.4.3. Ceiling, including all faces of secondary steel, stiffeners, penetrations, etc. - White (MUNSELL notation N 9.5).

9.4.4. Bracings, columns, void spaces, trunks, cofferdams - White (MUNSELL notation N 9.5).

9.4.5. Column Shaft, Lift shaft, Pontoon Tunnel - White (MUNSELL notation N 9.5).

9.4.6. Decks: Ice Grey (MUNSELL notation N 8).

9.4.7. Outfitting like stairways and others, follows the same color coding for external structures.

9.5. Structural Tanks

9.5.1. Inner side of Potable Water, Industrial Water, Fuel, Ballast, Peak Tank: White (MUNSELL notation N 9.5).

9.5.2. Inner side of Cargo tanks, Slop tank: White (MUNSELL notation N 9.5)

9.6. Accommodation Modules

9.6.1. External and internal, including all corridors walls and ceiling: White (MUNSELL notation N 9.5).

9.6.2. Internal corridor deck, internal rooms deck: White (MUNSELL notation N 9.5).

9.6.3. The corridors shall be identified at each stairway front, doors accessing outside area and exits at both sides, according to Manual de Sinalização Petrobras.

9.6.4. The architectural items, linings colors, etc. shall be according to the specific Project document

10. COLOR DEFINITION FOR PROCESS, UTILITIES AND PRESSURE VESSELS

10.1. All vessels and non-structural tanks shall be white (N 9.5) for internal area and ice grey (N 8) for external area, except for hydrocarbon fluid process, which shall be aluminum.

10.2. All non-structural tanks shall be internally painted White (Munsell N 9.5). In case of other materials the internal color shall be light cream, ice grey or white.

Note: All vessel and tanks shall be identified with TAG and Function to comply with NR-13.

TABLE 2 - COLOR IDENTIFICATION FOR PROCESS AND UTILITIES EQUIPMENT

System	Equipment	Color
Oil process	Launcher/Receiver Pig	Aluminum
	Production Heater	Aluminum
	Production Separator	Aluminum
	Atmospheric Separator	Aluminum
	Electrostatic Treater (TO)	Aluminum
Gas Process	Scrubber / Knock Out Drum (Safety Gas)	Aluminum
	Fuel Gas Vessels	Aluminum
	Commutation Vessel to TG's	Aluminum
	Heat Exchangers	Aluminum
	Gas Filters	Aluminum
Dehydration Unit	Filters	Aluminum
	Mol Sieve Adsorber Vessels	Aluminum
	Heat Exchangers	Aluminum
	Glycol Contactor Unit	Aluminum
	Glycol Regeneration Unit	Aluminum
	Other Glycol Vessels	Aluminum
	Glycol Vent Trap	Safety orange
	Glycol tank	Safety Orange
H2S removal	Depurator vessel	Aluminum
	Filters	Aluminum
	H2S removal vessel	Aluminum
	Amie contractor tower	Aluminum
CO2 Removal	Depurator vessel	Aluminum
	Filters	Aluminum
	Heat Exchangers	Aluminum
	Membrane unit	Aluminum
Water Injection	De-aerator (gas)	Aluminum
	De-aerator (Vacuum) Filters	internal: white (N 9.5) external: ice grey (N 8)
Water Treatment	Fresh Water Generator [RC]	internal: white (N 9.5) external: ice grey (N 8)
	Hydrophoric Tank	internal: white (N 9.5) external: ice grey (N 8)
	Plate Exchanger	internal: white (N 9.5) external: ice grey (N 8)
	Tank for Acid	Safety orange

	Tank for Hypochlorite	Safety orange
Industrial Water	Expansion Vessel (Pressurized with air)	Cinza Gelo
	Chemical Products Tank	Safety orange
	Filter	ice grey (N 8)
	Furnace and Boiler	ice grey (N 8)
	Water heater	ice grey (N 8)
	Chemical Injection	Safety orange
	Expansion Vessel (Gas Pressurized)	Aluminum
Diesel fuel	Diesel Filters	Aluminum
	Diesel Tank (internal & external)	Aluminum
	Centrifugal Purifier [RC]	Verde Pastel
Compressed Air	Service Air Vessel	ice grey (N 8)
	Air Dryer [RC]	ice grey (N 8)
Lubricating Oil	Oil Filter	Aluminum
Produced Water	Oily Water tank – External (internal white)	Aluminum
	Slop Tank	Aluminum
	Hydrocyclone	Aluminum
	Produced Water Flotator	Aluminum
	Oil-water separator	Aluminum
Flare System	Preassure Vessel	Aluminum
Hydraulic System	Hydraulic Fluid Accumulator [RC]	ice grey (N 8)
Compression systems	Pressure Vessel	Aluminum
	Filter	Aluminum
	Heat exchanger	Aluminum
Sulphate removal unit	Filters	Aluminum
	Membrane	Aluminum
	Cleaning tank	Aluminum
	Cleaning heater	Aluminum
	Chemical tanks	Safety orange
Dew Point Adjustment (JT)	Pressure Vessel	Aluminum
	Filter	Aluminum
Dew Point Adjustment (Cooling)	Pressure Vessel	Aluminum
	Heat exchanger	Aluminum
	Compressor	Aluminum
	Filter	Aluminum
Nitrogen Generation System	Pressure Vessel	ice grey (N 8)
	Membrane Unit	ice grey (N 8)
Chemical unit	Chemical tanks	Safety orange
Non-flammable fluid (general)	Pressure vessel and heat exchanger	internal: white (N 9.5) external: ice grey (N 8)

11. COLOR DEFINITION FOR PIPINGS, VALVES AND FITTINGS

11.1. All piping shall be top color painted white (N1) for internal areas, and ice grey (N8) for external areas, except for hydrocarbon piping which shall be aluminum and fire fighting piping which shall be painted Safety Red in all their extension.

11.2. The piping shall be identified by TAG and stripes to identify the fluid. See Figure 1.

11.3. The stripes for identification shall have 400mm width.

11.4. Alternatively to painting, in areas not subjected to oil spills a weather- and UV-resistant adhesive tape, resistant to fluid temperature, may be used for piping markings.

11.5. All piping inside tanks shall be identified with TAG and painted stripes. Adhesive tapes shall not be applied.

11.6. The identification stripe which includes two color is split in three sections, in a way that exist a 2:1 ratio for the central section which identifies the basic color or primary system, and external sections intended to additional secondary function that indicate the specific characteristic. See Figure 1.

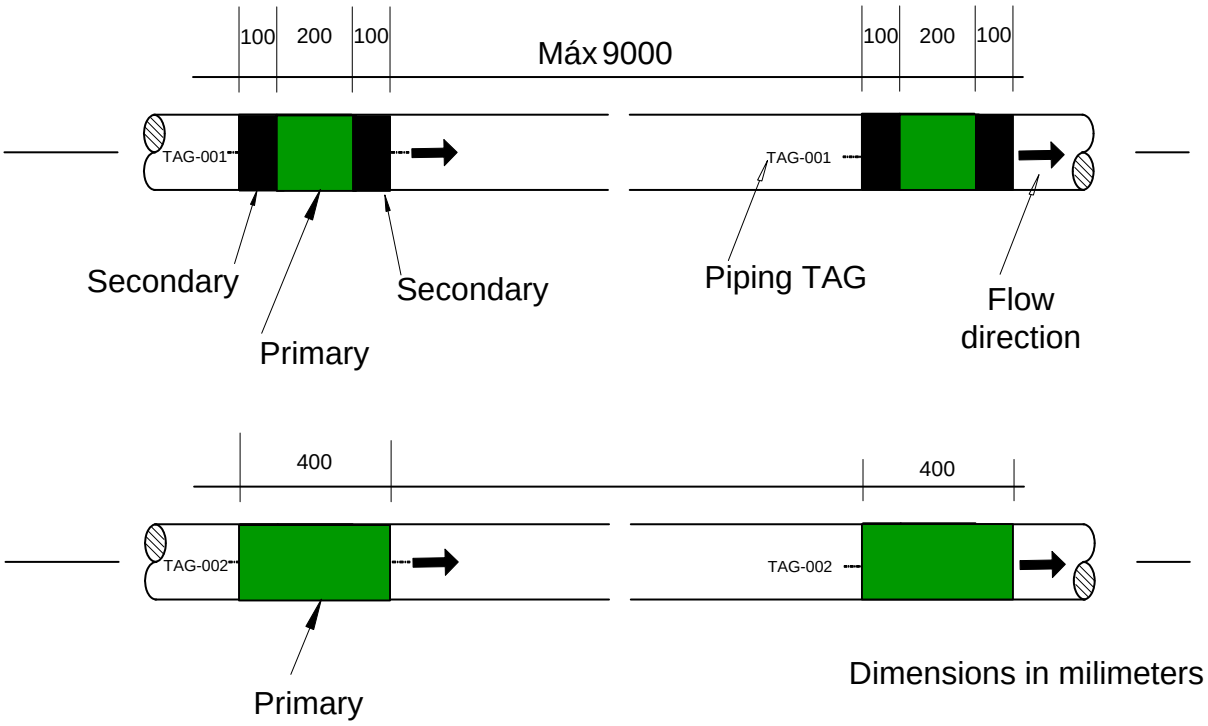

Figure 1 – Identification Stripes

11.7. All piping shall be painted or have stripes placed in such a way that the working staff may see and identify the piping and the product that it contains easily visible from the deck, platform or access walkway.

11.8. These stripes and flow direction arrows shall be placed each 9 meters length maximum, or wherever necessary (see Figure 2) for identification at:

11.8.1. Branches;

11.8.2. Close to all control valves/maneuver equipment, section isolation valves and supply point, hose stations, receiving ends;

11.8.3. Close to inlets and outlets of the equipment (suction and discharge lines at pumps, compressors, etc.);

11.8.4. Inlet and outlet of process train (vessels, tanks, etc.);

11.8.5. On each side of bulkhead or deck penetration points;

11.8.6. At the start and end of pipe racks; and

11.8.7. On elevated pipes, markers shall be positioned adjacent to stairways and platforms.

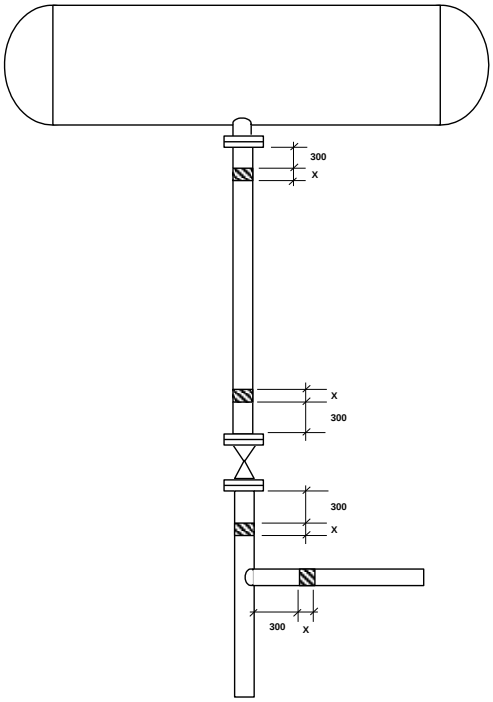

Figure 2 – Stripes location

11.9. All stripes colors shall be in accordance with the specified colors of the Table 3 (with adhesive tape or painted).

11.10. Line tags and arrows to indicate flow direction shall be painted (or marked by tape in internal areas), not necessarily inside the stripes (Figure 1). Text height shall be about 20 mm for ½”~ 3” piping diameters and about 50 mm for piping diameters above 3” diameters.

11.11. Marking color shall be as described below, ensuring contrast with the background.

11.11.1. Light colored piping: Black arrow/TAG (Munsell N 1)

11.11.2. Dark colored piping: White arrow/TAG (Munsell N 9.5)

11.12. All the valves and piping accessories (including filters, PSV, ADV, etc.) shall be painted with the same color of the piping.

11.13. Piping with thermal insulation do not require identification under the insulation but shall be identified by stripes and/or line TAG over the insulation.

11.14. On insulated piping with hidden electrical heat tracing, an additional warning label shall be added in Safety Yellow, with the words “Cuidado – Heat Tracing Elétrico”. Letters shall be Black.

11.15. The piping TAGS shall be according to Petrobras DR-ENGP-I.1.1 (Piping Standard and Material for Oil Production and Process Facilities) and Project P&ID’s.

11.16. All piping shall be painted with identification strips, according to Table 3 below, except for FRP, where there is no basic color requirement.

TABLE 3 - COLOR IDENTIFICATION FOR PIPING

Simbol	Service	Basic Color	Secundary stripe	Primary Stripe	Secundary stripe
ADW	Água de diluição Dilution water (Desalinated and aerated sea water)	internal: white external: ice grey	Green emblem	Green emblem	Green emblem
AM	Amina Amine	internal: white external: ice grey	Safety orange	Safety orange	Safety orange
BG	Gás inerte Inert gas	internal: white external: ice grey	Ligth grey	Safety yellow	Ligth grey
BR	Esgotamento Bilge	internal: white external: ice grey	Brown	Green emblem	Brown
BW	Lastro Ballast	internal: white external: ice grey	Green emblem	Green emblem	Green emblem
CHW	Água de resfriamento Chilled water	internal: white external: ice grey	Green emblem	Green emblem	Green emblem
CI	Injeção química não corrosiva Chemical injection (non-corrosive)	internal: white external: ice grey	Safety orange	Safety orange	Safety orange

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Simbol	Service	Basic Color	Secundary stripe	Primary Stripe	Secundary stripe
CN	Hipoclorito de sódio Sodium Hypochlorite	internal: white external: ice grey	Safety orange	Safety orange	Safety orange
CNI	Injeção química corrosiva Corrosive chemical injection (except for sodium hypochlorite)	internal: white external: ice grey	Safety orange	Safety orange	Safety orange
D	Óleo diesel Diesel	Aluminum	Brown	Aluminum	Brown
DA	Dreno Aberto Open drain	internal: white external: ice grey	Brown	Green emblem	Brown
DF	Dreno fechado nãocorrosivo Closed drain (non-corrosive hydrocarbon)	Aluminum	Brown	Brown	Brown
DFC	Dreno fechado corrosivo Corrosive closed drain (corrosive hydrocarbon)	Aluminum	Brown	Brown	Brown
DW	Distilled water	internal: white external: ice grey	Green emblem	Green emblem	Green emblem
DO	Dióxido de carbono Carbon dioxide (Dry)	Safety red	Safety red	Safety red	Safety red
DOH	Dióxido de carbono com hidrocarboneto Carbon Dioxide w/ Hydrocarbon	Aluminum	Safety yellow	Safety yellow	Safety yellow
F	Flare e sistema de segurança Safety system and flare	Aluminum	Safety yellow	Safety yellow	Safety yellow
FG	Gás combustível Fuel gas	Aluminum	Safety yellow	Safety yellow	Safety yellow
FL	Espuma Foam line	Safety red	Safety red	Safety red	Safety red
FW	Água de incêndio Fire water (aerated sea water)	Safety red	Safety red	Safety red	Safety red
G	Glicol Glycol	internal: white external: ice grey	Safety orange	Safety orange	Safety orange
GL	Gás liquefeito de petróleo Refrigered liquified petroleum gases (Flare process, drain system)	Aluminum	Brown	Safety yellow	Brown
GW	Águas Cinzentas Grey water	internal: white external: ice grey	Brown	Green emblem	Brown
H	Óleo hidráulico Hidraulic fluid	Aluminum	Orange	Aluminum	Orange
HF	Óleo térmico Hot fluid	Aluminum	Orange	Aluminum	Orange
HWP	Água de aquecimento de processo Process heating water	internal: white external: ice grey	Green emblem	Green emblem	Green emblem

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Simbol	Service	Basic Color	Secundary stripe	Primary Stripe	Secundary stripe
HWU	Água de aquecimento de utilidades Utility heating water	internal: white external: ice grey	Green emblem	Green emblem	Green emblem
IA	Ar de instrumentação Instrumentation air	internal: white external: ice grey	Safety blue	Safety blue	Safety blue
IG	Gás de instrumentos Instrumentation gas	Aluminum	Safety yellow	Safety yellow	Safety yellow
IGN	Linha de ignição Ignition Line	Aluminum	Safety yellow	Safety yellow	Safety yellow
IN	Nitrogênio de Instrumentação Instrumentation Nitrogen	internal: white external: ice grey	Ligth grey	Safety yellow	Ligth grey
IW	Água de injeção desaerada De-areted seawater injection	internal: white external: ice grey	Green emblem	Green emblem	Green emblem
IWC	Água de injeção aerada Aerated sea injection water	internal: white external: ice grey	Green emblem	Green emblem	Green emblem
JF	Querosene de aviação Aviation kerosene	Aluminum	Brown	Aluminum	Brown
OL	Óleo lubrificante Lubrificating oil	Aluminum	Orange	Aluminum	Orange
P*	Hidrocarboneto não corrosivo Process (non-corrosive hydrocarbon) or Loading Line (Naval System)	Aluminum	Brown Safety yellow	Brown Safety yellow	Brown Safety yellow
PC*	Hidrocarboneto corrosivo Process (corrosive hydrocarbon) or slop unloading system (Naval System)	Aluminum	Brown Safety yellow	Brown Safety yellow	Brown Safety yellow
* For multiphase or liquid fluids, the color of the strip shall be Brown. If it is in the gas phase, the stripe shall be Safety yellow.					
PCW	Água produzida Produced water	Aluminum	Brown	Green emblem	Brown
PW	Água potável Potable water	internal: white external: ice grey	Green emblem	Green emblem	Green emblem
PWH	Água potável quente Hot fresh water	internal: white external: ice grey	Green emblem	Green emblem	Green emblem
S	Vapor Steam	internal: white external: ice grey	White	White	White
SA	Ar de serviço geral Service compressed air (utility air)	internal: white external: ice grey	Safety blue	Safety blue	Safety blue
SC	Vapor Condensado Steam condensed	internal: white external: ice grey	White	Green emblem	White
SFW	Água doce de incêndio Fresh fire fighting water	Safety red	Safety red	Safety red	Safety red
SN	Nitrogênio para partida	internal: white	Ligth grey	Safety yellow	Ligth grey

Simbol	Service	Basic Color	Secundary stripe	Primary Stripe	Secundary stripe
	Start-up nitrogen	external: ice grey			
SP	Vent and tank sounding	Deve seguir a cor do fluido do tanque de referência.			
SW	Água salgada/ resfriamento Aerated sea water	internal: white external: ice grey	Green emblem	Green emblem	Green emblem
SWW	Água salgada de lavagem Aerated sea flushing water (Naval System)	internal: white external: ice grey	Green emblem	Green emblem	Green emblem
W	Água industrial / resfriamento Industrial water (fresh, non-corrosive water or deaerated sea water)	internal: white external: ice grey	Green emblem	Green emblem	Green emblem
WW	Águas negras Sewage waste water	internal: white external: ice grey	Brown	Green emblem	Brown

12. COLOR DEFINITION FOR MARINE SYSTEMS

12.1. Mooring winches [RC]: Safety Yellow (MUNSELL notation 5 Y 8/12).

12.2. Fairleader: Follow the adjacent structure.

12.3. Pull-in winch: Safety Yellow (MUNSELL notation 5 Y 8/12).

12.4. Vacuum Unit [RC]: Light Green (MUNSELL notation 5 G 8/4).

12.5. Vent Post: Aluminum (without MUNSELL notation).

13. COLOR DEFINITION FOR MECHANICAL, CARGO HANDLING AND PACKAGE EQUIPMENT

13.1. Pumps in General (RC): Light Green (MUNSELL notation 5 G 8/4), except fire-fighting system (see item 4.6).

13.2. Air compressors [RC], Air Dryers [RC]: Light Green (MUNSELL notation 5 G 8/4

13.3. Turbo-machinery hood and machinery in general, compressors [RC] and turbines hoods [RC]: Ice Gray (MUNSELL notation N 8).

13.4. Acoustic Hood for engines: According to compartment color.

13.4.1. Internal White (MUNSELL notation N 9.5).

13.4.2. External Ice Gray (MUNSELL notation N 8).

13.5. Diesel engines [RC]: Light Green (MUNSELL notation 5 G 8/4).

13.6. Boilers and Furnace: Aluminum (without MUNSELL notation).

13.7. Exhaust gas pipe/ducts for turbine, engines, boiler, furnace: Aluminum (without MUNSELL notation).

13.8. Hydraulic Unit [RC]: Light Green (MUNSELL notation 5 G 8/4).

13.9. Diesel Oil Centrifugal Purifier [RC]: Light Green (MUNSELL notation 5 G 8/4

13.10. Cranes: Safety Yellow (MUNSELL notation 5 Y 8/12).

13.11. Equipment and Package Unit Skid Beam: follows the same color of base structure or deck where it is mounted on)

13.12. Hoisting Devices (monorails, frames, trolleys, trolley beams, lifting lugs, etc.): Safety Yellow (MUNSELL notation 5 Y 8/12).

13.13. Guards and Protective Cover for dangerous moving, rotating parts of Equipment & Machines - Safety Orange (MUNSELL notation 2.5 Y R 6/14).

13.14. Dangerous moving, rotating exposed parts of machinery: Safety Orange (MUNSELL notation 2.5 Y R 6/14).

13.15. External face of Pulleys and Exposed Gears: Safety Orange (MUNSELL notation 2.5 Y R 6/14).

13.16. Potable Water Maker [RC]: Light Green (MUNSELL notation 5 G 8/4).

13.17. Sewage Treatment Unit [RC]: Light Green (MUNSELL notation 5 G 8/4).

14. COLOR DEFINITION FOR SAFETY

14.1. Fire-fighting Pumps and Drives: Safety Red (MUNSELL notation 5 R 4/14).

14.2. Fire-fighting Systems Equipment in general: Safety Red (MUNSELL notation 5 R 4/14).

14.3. Fire-fighting Valves and Piping Accessories, Sprinkler: Safety Red (MUNSELL notation 5 R 4/14).

14.4. Foam Tank: Safety Red (MUNSELL notation 5 R 4/14).

14.5. Carbon Dioxide: CO_2 Bottle, piping and diffuser – Safety Red (MUNSELL notation 5 R 4/14).

14.6. Portable fire extinguisher: Safety Red (MUNSELL notation 5 R 4/14).

14.7. Life-Saving: Safety Orange (MUNSELL notation 2.5 YR 6/14).

14.8. Lifeboat:

14.8.1. External: Safety Orange (MUNSELL notation 2.5 YR 6/14).

14.8.2. Internal: White (MUNSELL notation N 9.5).

14.9. Rescue Boat: Safety Orange (MUNSELL notation 2.5 YR 6/14).

14.10. First aid equipment box: PETROBRAS Green (MUNSELL notation 2.5 G 5/10).

14.11. Breathing apparatus lockers: PETROBRAS Green (MUNSELL notation 2.5 G 5/10).

14.12. Safety Showers and Stretchers, Eye-wash Unit: PETROBRAS Green (MUNSELL notation 2.5 G 5/10).

14.13. Escape Route shall be painted in PETROBRAS Green (MUNSELL notation 2.5 G 5/10) in anti-slippery coating, surrounded on each side by 100 mm width stripes and having arrows spaced at maximum 3000 mm. Arrows and stripes shall be in White (MUNSELL notation N 9.5).

14.14. Emergency exit doors: Safety-Red (MUNSELL notation 5 R 4/14) with marking “Saída de Emergência”, “Emergency Exit” (red letters) over horizontal 500mm height White stripe.

14.15. Escape Hatches: Safety-Red (MUNSELL notation 5 R 4/14).

14.16. Safety hatches: Safety net (MUNSELL Safety net 5 R 4/14).

14.17. Wall at the end of no-way out corridors; Safety Yellow (MUNSELL notation 5 8/12).

14.18. Hospital entrance door: PETROBRAS Green (MUNSELL notation 2.5 G 5/10).

14.19. Area on the deck surrounding safety equipment, firefighting equipment or other emergency equipment: White (MUNSELL notation N 9.5).

14.20. Area on the deck surrounding electric switchgear/panels shall be signalized by painting a 100mm yellow stripe in the electric arc protection area.

15. COLOR DEFINITION FOR HVAC

15.1. Ventilating & Exhaust Fans: According to compartment color.

15.1.1. White (MUNSELL notation N 9.5).when located internal side; and

15.1.2. Ice Gray (MUNSELL notation N 8) when located outdoor.

15.2. Fan Mushrooms, Dampers, Filter frame: Ice Gray (MUNSELL notation N 8).

Note: Watertight, Fire tight dampers handles for manual operation shall be Safety Orange (MUNSELL notation 2.5 Y R 6/14).

15.3. Ducts:

15.3.1. Internal White (MUNSELL notation N 9.5)

15.3.2. External: Ice Gray (MUNSELL notation N 8).

15.4. Ar conditioning package unit, Chillers: Green (MUNSELL notation 5 G 8/4).

16. COLOR DEFINITION FOR ELETRIC EQUIPMENT

16.1. Generator (RC): Light Green (MUNSELL notation 5 G 8/4).

16.2. Motor (RC): Light Green (MUNSELL notation 5 G 8/4), except fire-fighting pump and small motors assembled in Package Units).

16.3. Transformer enclosure (RC): Light Green (MUNSELL notation 5 G 8/4).

16.4. Electric Switchboard, Distribution Board Enclosure (RC): Light Green (MUNSELL notation 5 G 8/4).

16.5. UPS [RC], Converters [RC], Rectifiers [RC]: Light Green (MUNSELL notation 5 G 8/4).

16.6. Control Gear [RC], Control Panel for Package Equipment [RC]: Light Green (MUNSELL notation 5 G 8/4). *See Notes below.

*Note: Inner components mounting plate and/or inner door/safety barrier of Electrical Panels referred above: Safety Orange (MUNSELL notation 2.5 Y R 6/14)

16.7. Cable Tray, Steel Conduit, Junction Box:

16.7.1. Metallic: According to compartment color

16.7.2. FRP: no requirement of color.

16.8. Lighting Fixtures:

16.8.1. Carbon Steel Parts [RC]: White (MUNSELL Notation N 9.5).

16.8.2. FRP or stainless steel: no requirement of color

17. COLOR DEFINITION FOR INSTRUMENTATION

17.1. Instruments: follow the piping or equipment color. * See note bellow. Instrumentos:

17.2. Control Panels [RC]: Light Cream (MUNSELL notation 2.5 Y 9/4).

17.3. Control valves and actuators: follow the piping color. * See note bellow.

* Note: For new parts, it may be kept the original maker painting, except for control valves.

18. COLOR DEFINITION FOR TELECOMUNICATION

18.1. Telecommunication Tower: In accordance with Brazilian Administration Requirements (Portaria nº 0256/GC5).

19. COLOR DEFINITION FOR HELIDECK LANDING AREA

19.1. According to NORMAM 27.

20. COLOR DEFINITION FOR MISCELLANEOUS

20.1. Stripe at the elevators entrance floor: Safety Yellow (MUNSELL notation 5 Y 8/12)

20.2. SOPEP KIT box: Safety Orange (MUNSELL notation 2.5 Y R 6/14) with marking “KIT SOPEP” (black letters)